

Dynamic PON Monitoring System Based on Time-Frequency Encoded Optical Pulses

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Abstract We propose a novel PON monitoring method using time–frequency encoded optical pulse sequences and cross-correlation analysis for real-time fault localization and port detection in dynamic port switching scenarios. Experimental results show that the proposed method achieves sub-meter localization accuracy and 100% port identification accuracy. ©2026 The Author(s)

Introduction

Driven by the growing demand for diverse broadband services, access networks have expanded rapidly. As a critical bridge between subscribers and the backbone network, passive optical networks (PONs) face increasingly prominent challenges in operation and maintenance [1–2]. Optical time domain reflectometry (OTDR) enables fiber condition monitoring by analyzing backscattered signals. However, due to the lack of identity information for each port in PONs, OTDR struggles to distinguish events occurring on different branches. To enable intelligent operation and maintenance of PONs, a variety of monitoring schemes that utilize port coding have emerged.

Fiber bragg grating (FBG) technology has attracted wide attention in port encoding due to its high immunity to electromagnetic interference and excellent multiplexing capability. Zhang et al. [3] used Fabry–Pérot cavity resonance for generating distinct spectral signatures. Fernández et al. [4] employed dispersive Fourier transformation to map FBG spectral features into the time domain, enabling the identification of branch links. Zhang et al. [5] proposed a WBG-based remote coding scheme in PLC splitters for PON monitoring, enabling identification of equidistant users. Existing coding schemes perform well in static optical network environments. However, the flexibility and adaptability of these schemes remain limited in dynamic scenarios where the mapping between optical network units (ONU) and ports changes frequently due to factors such as fault recovery, network reconfiguration, and load balancing. A more flexible and adaptive coding approach is thus essential for future intelligent optical access networks.

In our previous work, we have proposed a FBGs based ports encoders to achieve the 16-branch PON smart monitoring [6]. However, this system is time-consuming and cannot achieve automatic monitoring of port switching. In this paper, we have proposed a time-frequency division coding method based on a cascaded port coding scheme using FBGs to enable dynamic port

monitoring, fiber link fault detection, and PON supervision. By employing this approach, we achieve millisecond-level monitoring response and sub-meter localization accuracy (within 1 m), and verify its performance in a 2×4 PON over a 1 km distance under dynamic port-switching scenarios. Compared with the conventional FBG cascade encoding method using a tunable laser OTDR (TL-OTDR), the proposed method improves detection speed by a factor of 6.

The Working Principle

The proposed dynamic PON monitoring system is based on time–frequency encoded optical pulse sequences and cross-correlation analysis, in which the initial pulse sequence is filtered by a wavelength-selective encoder to assign a unique code to each port, propagates through the fiber link, and is reflected back from either ONU reflectors or fault points. The returned signals are applied with cross-correlation analysis to distinguish normal ports (exhibiting maximum correlation peaks indicative of link length) from faulty ones (exhibiting attenuated reflector peaks but sharp event reflections). The method offers dynamic compatibility for port-switching scenarios, high-precision localization through noise-resistant correlation processing, and wavelength-multiplexed multi-channel diagnosis.

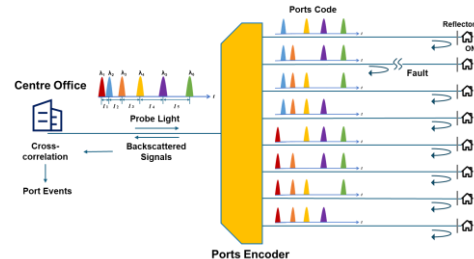


Fig. 1: The proposed dynamic port–link monitoring system.

To improve demodulation accuracy, the generated optical pulse sequences must possess distinct autocorrelation features. Conventional uniformly spaced configurations typically produce suboptimal autocorrelation profiles with prominent interference peaks, resulting in elevated

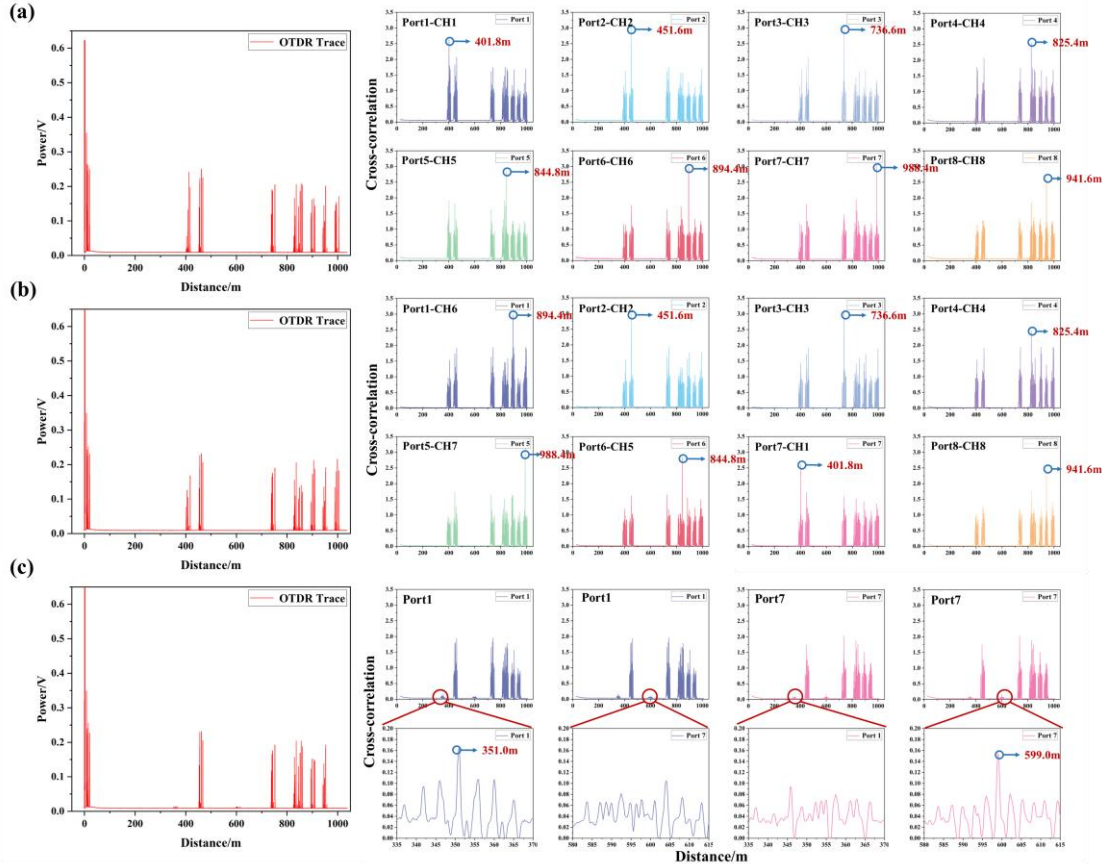


Fig. 4: OTDR trace and port code cross-correlation detection results for (a) Test 1; (b) Test 2; (c) Test 3

matches. The improved scheme demonstrates 25% enhancement in ESR, increasing from 1.2 to 1.5, thereby ensuring superior noise immunity, and maintained real-time processing capability, which is presented in Fig. 3.

Furthermore, we have tested dynamic port switching performance to compare the cross-correlation demodulation results of Test 1 (pre-switch) and Test 2 (port-switch), see in Fig. 4(a) and (b). The experiment results showed a sub-meter localization accuracy (± 1 m) validated against reference lengths (in Tab.1) and 100% identification of all port-link connection states. After switching the connections of ports 1, 5, 6, and 7, the system successfully tracked the ONU connection changes, confirming its effectiveness in real-time tracking and identification of port-link connectivity in PONs under dynamic switching conditions.

To further evaluate the system's capability for simultaneous detection and localization of multiple events, the port 1 and port 7 were with the artificial faults at approximately 350 m and 600 m during Test 3. Fig. 4 (c) presents the cross-correlation demodulation results for the encoded signals of ports 1 and 7. The experimental results reveal that grating reflection signals were absent at the fault ports, whereas the Fresnel reflection signals of artificial faults exhibited strong

correlation with the corresponding port encoding signals, characterized by pronounced peak intensity and sharpness. In contrast, the demodulated signals of non-fault ports showed no interference peaks at the fault locations. Therefore, the proposed system achieved accurate identification of multiple events, including port switching and connection faults, with 100% port recognition and <1 m localization error.

Conclusions

This paper presented a novel PON monitoring method employing a time-frequency encoded optical pulses sequence and cascaded FBG filters for port coding and event detection. The proposed architecture achieves three key advancements through a single broadband optical scan: millisecond-timescale port-link connectivity detection, unique event encoding in dynamic PON switching environments, and concurrent identification of multiple network events. Against conventional TL-OTDR based FBG cascade systems, the proposed system demonstrates significant performance gains, including a 6 times enhancement in detection speed and preserving sub-meter fault localization accuracy. These experimental results validate the technical feasibility of the proposed system for real-time, high-throughput monitoring in next-generation intelligent optical access networks.

Acknowledgements

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